

In [26]: `!pip install statsmodels`

```
Requirement already satisfied: statsmodels in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (0.15.0)
Requirement already satisfied: numpy<3,>=1.23.5 in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from statsmodels) (2.4.4)
Requirement already satisfied: scipy!=1.9.2,>=1.8 in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from statsmodels) (1.18.1)
Requirement already satisfied: pandas!=2.1.0,>=1.4 in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from statsmodels) (3.0.2)
Requirement already satisfied: patsy>=0.5.6 in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from statsmodels) (1.0.3)
Requirement already satisfied: packaging>=21.3 in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from statsmodels) (26.0)
Requirement already satisfied: formulaic>=1.1.0 in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from statsmodels) (1.2.2)
Requirement already satisfied: interface-meta>=1.2.0 in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from formulaic>=1.1.0->statsmodels) (2.0.1)
Requirement already satisfied: narwhals>=1.17 in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from formulaic>=1.1.0->statsmodels) (2.26.0)
Requirement already satisfied: typing-extensions>=4.2.0 in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from formulaic>=1.1.0->statsmodels) (4.15.0)
Requirement already satisfied: wrapt>=1.17.0rc1 in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from formulaic>=1.1.0->statsmodels) (2.4.1)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pandas!=2.1.0,>=1.4->statsmodels) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pandas!=2.1.0,>=1.4->statsmodels) (2026.1)
Requirement already satisfied: six>=1.5 in C:\Users\Admin\AppData\Local\Programs\Python\Python313\Lib\site-packages (from python-dateutil>=2.8.2->pandas!=2.1.0,>=1.4->statsmodels) (1.17.0)
[notice] A new release of pip is available: 26.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [27]: `import pandas as pd`
`import seaborn as sns`
`import matplotlib.pyplot as plt`

In [28]: `#Task 1 – Instagram Likes vs Comments`
`df = pd.read_csv("instagram_likes_comments.csv")`

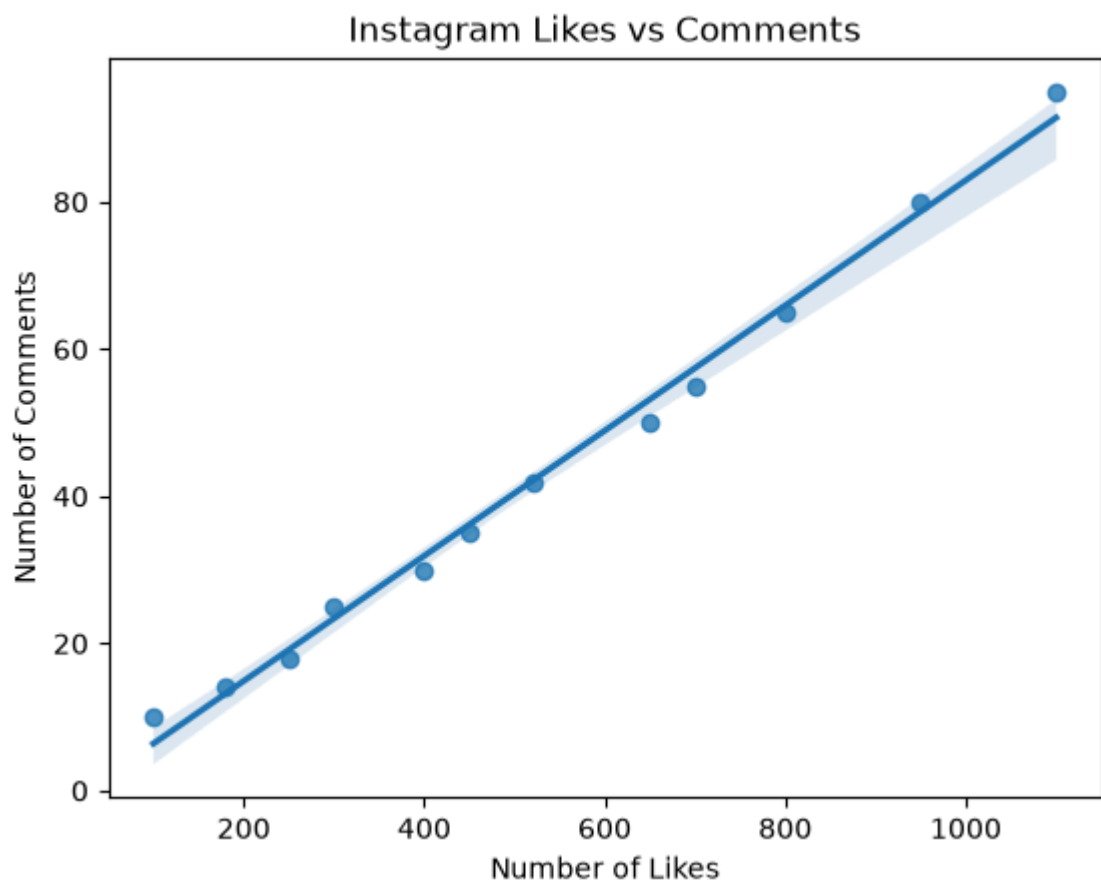
In [29]: `df`

Out[29]:

	likes	comments
0	100	10
1	250	18
2	180	14
3	400	30
4	520	42
5	650	50
6	300	25
7	800	65
8	950	80
9	700	55
10	1100	95
11	450	35

```
In [30]: sns.regplot(x="likes", y="comments", data=df)
plt.xlabel("Number of Likes")
plt.ylabel("Number of Comments")
plt.title("Instagram Likes vs Comments")
```

Out[30]: Text(0.5, 1.0, 'Instagram Likes vs Comments')



#Conclusion- The regression plot shows a positive relationship between likes and comments. As the number of likes increases, the number of comments generally increases.

```
In [31]: #Task 2 Task 2 – Zomato Price vs Delivery Time
```

```
df = pd.read_csv("zomato_price_delivery.csv")
```

```
In [32]: df
```

```
Out[32]:
```

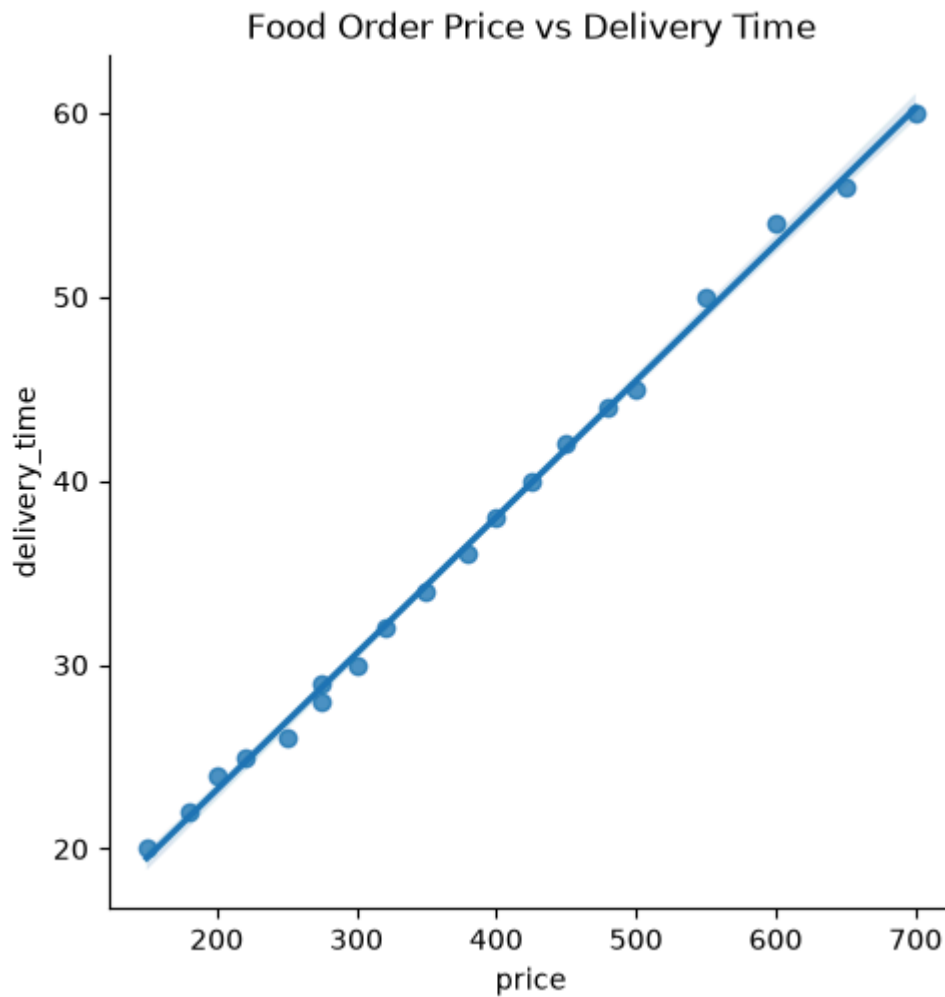
	price	delivery_time
--	-------	---------------

0	150	20
1	200	24
2	250	26
3	180	22
4	300	30
5	350	34
6	400	38
7	220	25
8	275	29
9	450	42
10	500	45
11	320	32
12	380	36
13	550	50
14	600	54
15	425	40
16	700	60
17	275	28
18	480	44
19	650	56

```
In [33]: sns.lmplot( x="price", y="delivery_time", data=df)
```

```
plt.title("Food Order Price vs Delivery Time")
```

```
Out[33]: Text(0.5, 1.0, 'Food Order Price vs Delivery Time')
```



#Conclusion The Implot shows the relationship between food order price and delivery time. In this sample data, higher order prices are generally associated with longer delivery times.

In [4]: *#Task 3 – Spotify Playlist: Lowess Smoothing*

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
```

In [5]: `df = pd.read_csv("spotify_playlist(1).csv")`

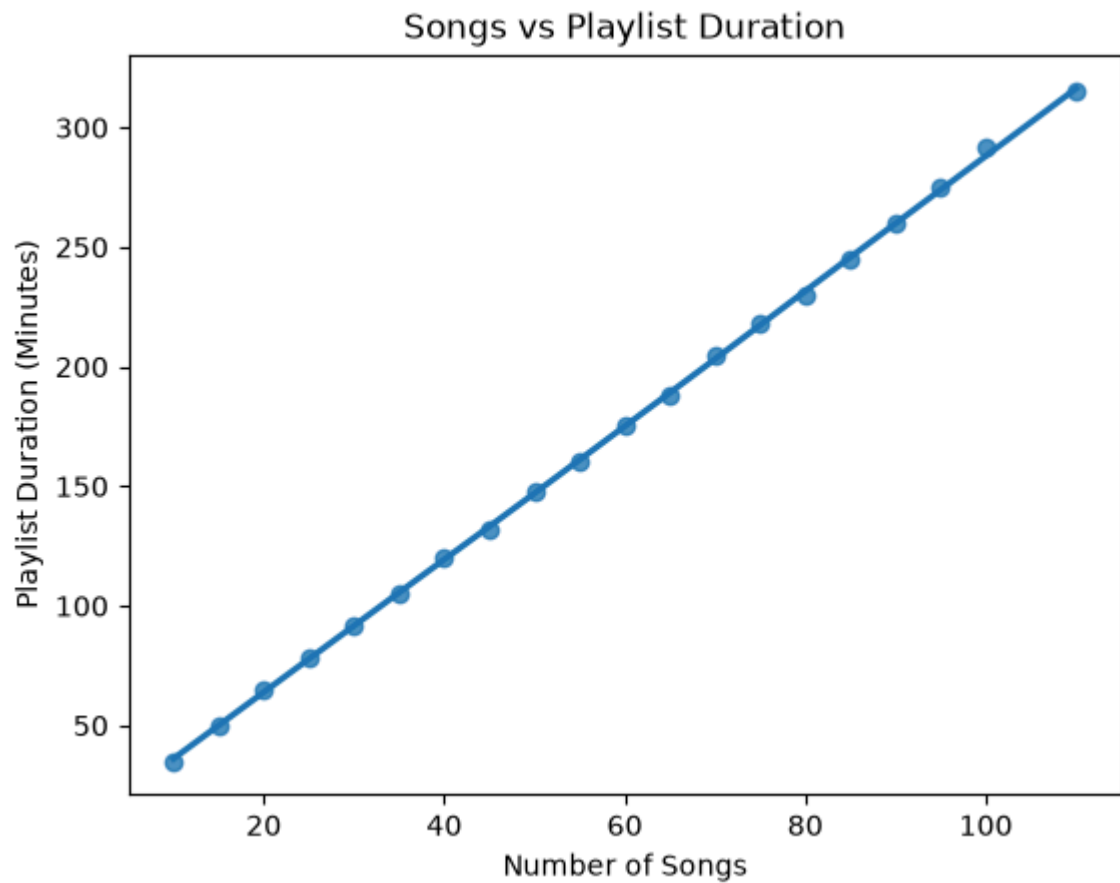
In [6]: `df`

Out[6]:

	songs	duration
0	10	35
1	15	50
2	20	65
3	25	78
4	30	92
5	35	105
6	40	120
7	45	132
8	50	148
9	55	160
10	60	175
11	65	188
12	70	205
13	75	218
14	80	230
15	85	245
16	90	260
17	95	275
18	100	292
19	110	315

```
In [7]: sns.regplot( x="songs", y="duration", data=df, lowess = True)
plt.xlabel("Number of Songs")
plt.ylabel("Playlist Duration (Minutes)")
plt.title("Songs vs Playlist Duration")
```

```
Out[7]: Text(0.5, 1.0, 'Songs vs Playlist Duration')
```



#Conclusion The LOWESS curve shows the relationship between the number of songs and total playlist duration. As the number of songs increases, the playlist duration also increases.

```
In [8]: #Task 4 – Faceted Regression Plot  
  
df = pd.read_csv("youtube_exam_preparation.csv")
```

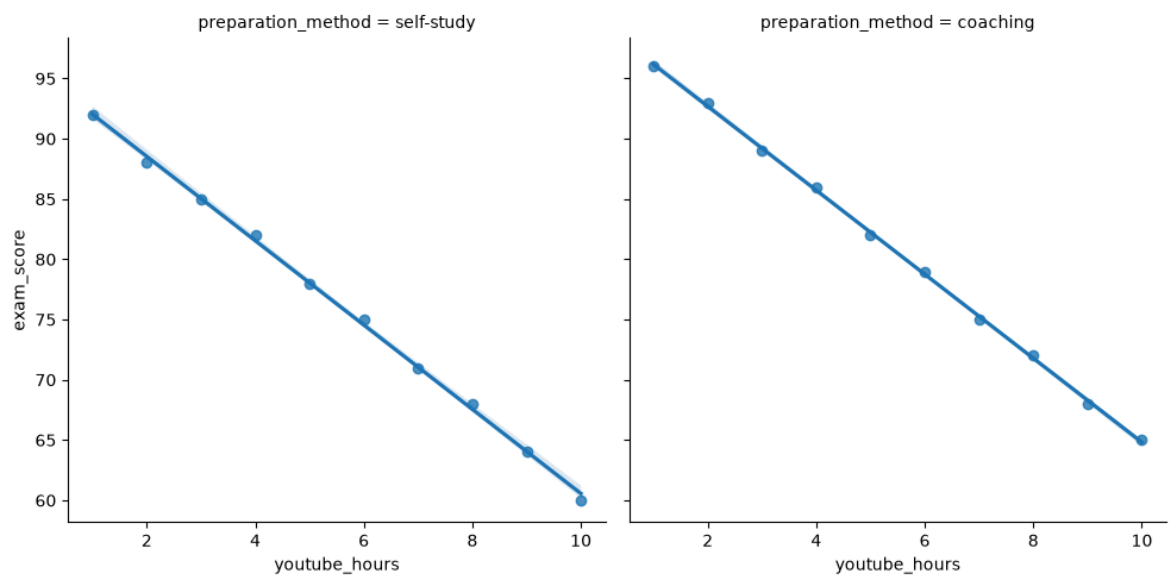
```
In [9]: df
```

Out[9]:

	youtube_hours	exam_score	preparation_method
0	1	92	self-study
1	2	88	self-study
2	3	85	self-study
3	4	82	self-study
4	5	78	self-study
5	6	75	self-study
6	7	71	self-study
7	8	68	self-study
8	9	64	self-study
9	10	60	self-study
10	1	96	coaching
11	2	93	coaching
12	3	89	coaching
13	4	86	coaching
14	5	82	coaching
15	6	79	coaching
16	7	75	coaching
17	8	72	coaching
18	9	68	coaching
19	10	65	coaching

```
In [10]: sns.lmplot( x="youtube_hours", y="exam_score", col="preparation_method", data=df
```

```
Out[10]: <seaborn.axisgrid.FacetGrid at 0x258040d5010>
```



#Conclusion The faceted regression plot compares YouTube hours and exam scores for students using self-study and coaching. In both groups, the sample data shows a negative relationship: as YouTube hours increase, exam scores generally decrease.

```
In [11]: #Task 5 – Flipkart Price vs Reviews

df = pd.read_csv("flipkart_price_reviews.csv")
```

```
In [12]: df
```

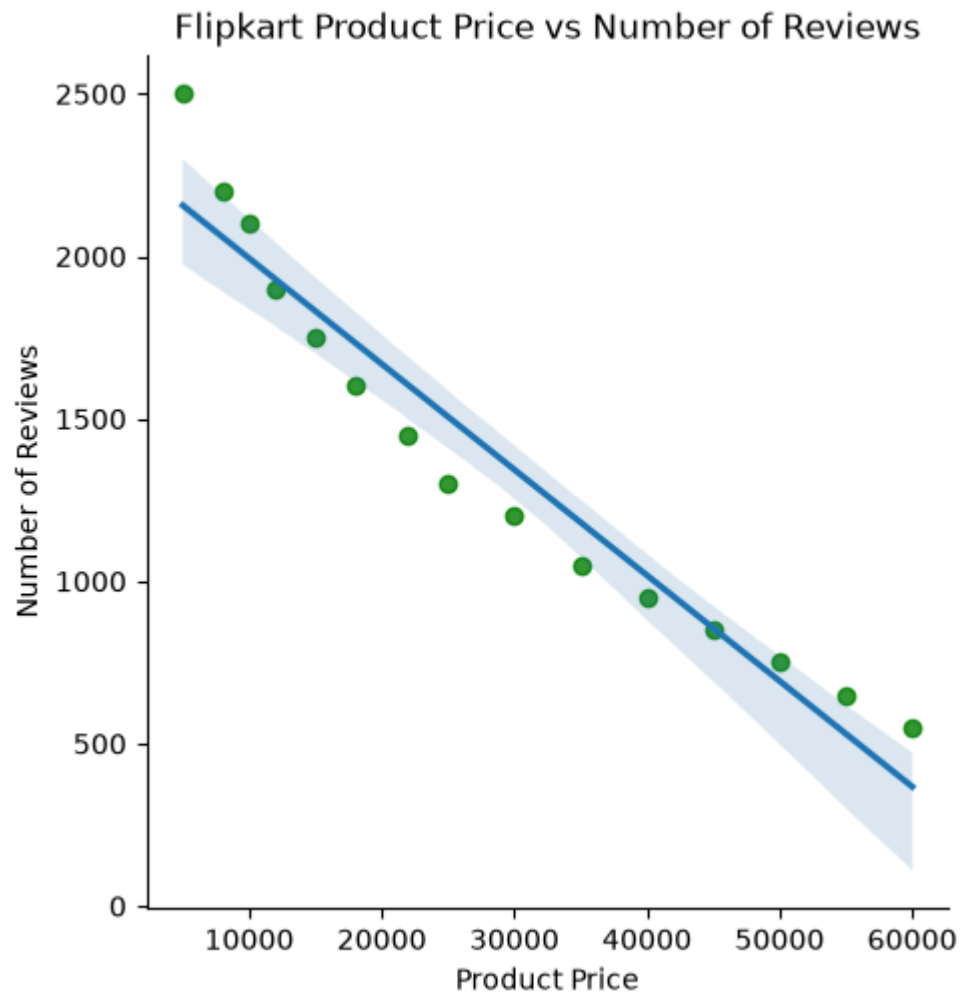
```
Out[12]:
```

	price	reviews
0	5000	2500
1	8000	2200
2	10000	2100
3	12000	1900
4	15000	1750
5	18000	1600
6	22000	1450
7	25000	1300
8	30000	1200
9	35000	1050
10	40000	950
11	45000	850
12	50000	750
13	55000	650
14	60000	550

```
In [13]: sns.lmplot( x="price", y="reviews", data=df, scatter_kws={"color": "green"})

plt.title("Flipkart Product Price vs Number of Reviews")
plt.xlabel("Product Price")
plt.ylabel("Number of Reviews")
```

```
Out[13]: Text(15.33658854166668, 0.5, 'Number of Reviews')
```



#Conclusion The regression plot shows the relationship between product price and number of reviews. The plot has been customized by adding a title, axis labels, and changing the marker color without changing the regression logic.